A Variable Definitions

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

A.1 Sample Manifest

Name	Description / Formula	Source	Reference
`file_name`, `ceo_id`, `ceo_name`, `gvkey`, `ff12_code`, `ff12_name`, `start_date`	Master sample manifest with CEO, firm, and industry identifiers	outputs/1.4_AssembleManifest	

A.2 Text/Linguistic Variables

Name	Description / Formula	Source	Reference
`NegCall`	Entire-call negative sentiment percentage. *Formula:* Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: 'percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.'	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.4, p.18; Appendix Table A.1, p.54)
`UncAnsCEO`	CEO-only Q&A uncertainty percentage. *Formula:* CEO Q&A uncertainty (%) = UnctWordsAnsCEO / WordsAnsCEO (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary 'uncertainty' list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 2, p.13; Appendix Table A.1, p.54)
`UncAnsMgr`	Manager Q&A uncertainty percentage (all managers including CEO). *Formula:* Manager Q&A uncertainty (%) = (count of LM 2011 uncertainty-wordlist words spoken by managers in Q&A segment) / (total words spoken by managers in Q&A segment). Construction strategy follows DWZ 2021 Eqn (2); manager-pool identification follows BGT 2018 segment-split code (iangow/bgt replication). 'Manager' pool includes CEO plus other managerial speakers, extending DWZ which separates CEO/CFO.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (measurement strategy, Eqn 2); Bushee et al. (2018) (manager-pool identification via iangow/bgt); thesis extension to pooled manager speakers
`UncPreCEO`	CEO-only presentation uncertainty percentage. *Formula:* CEO Presentation uncertainty (%) = UnctWordsPreCEO / WordsPreCEO (DWZ 2021 Eqn 1). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 1, p.13; Appendix Table A.1, p.54)
`UncPreMgr`	Manager presentation uncertainty percentage. *Formula:* Manager Presentation uncertainty (%) = (count of LM 2011 uncertainty-wordlist words by managers in Presentation segment) / (total manager words in Presentation segment). Construction strategy follows DWZ 2021 Eqn (1); manager-pool identification and Pres/Ans segment split follow BGT 2018 via iangow/bgt replication code.	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (measurement strategy, Eqn 1); Bushee et al. (2018) (segment split + manager pool); thesis extension to pooled manager speakers
`UncQue`	Analyst Q&A uncertainty percentage. *Formula:* Analyst Q&A uncertainty (%) = UnctWordsQue / WordsQue (DWZ 2021 Eqn 3). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	Dzielínski et al. (2021) (Section 4.2, Eqn 3, p.13; Appendix Table A.1, p.54)

A.3 Financial / Market Variables

Name	Description / Formula	Source	Reference
BGTAvg_Amihud	BGT 25-day symmetric average Amihud illiquidity (H7e). *Formula:* Mean daily Amihud illiquidity over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (symmetric average shape)
BGTAvg_Spread	BGT 25-day symmetric average bid-ask spread (H14e). *Formula:* Mean daily relative bid-ask spread over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). Spread = (ASK - BID) / ((ASK + BID) / 2). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (symmetric average shape)
BGTDelta_Amihud	BGT 25-day post-pre Amihud illiquidity change (H7d). *Formula:* Mean Amihud over post window [+1, +25] minus mean over pre window [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (delta shape)
BGTDelta_Spread	BGT 25-day post-pre bid-ask spread change (H14d). *Formula:* Mean spread over [+1, +25] minus mean over [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (delta shape)
BGTLevel_Amihud	Compute BGT (2018) long-window Amihud illiquidity around earnings calls. *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/bgt_long_window_amihud.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) 56(1):85-121
BGTLevel_Spread	Compute BGT (2018) long-window closing bid-ask spread around earnings calls. *Formula:* CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/f1d/shared/variables/bgt_long_window_spread.py)	CRSPviaget_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula]
Capex	Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data. *Formula:* Compustat/capxy_Q4/atq (see module: src/f1d/shared/variables/capex_intensity.py)	Compustat/capxy_Q4/atq	Bates, Kahle & Stulz (2009, JF)
CashFlowAt	Build CashFlow = oancfy / atq from raw Compustat quarterly data. *Formula:* Compustat/oancfy/atq (see module: src/f1d/shared/variables/cash_flow.py)	Compustat/oancfy/atq	Bates, Kahle & Stulz (2009, JF)
CashRatio	Build CashRatio = cheq / atq from raw Compustat quarterly data. *Formula:* Compustat/cheq/atq (see module: src/f1d/shared/variables/cash_holdings.py)	Compustat/cheq/atq	Bates et al. (2009) (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)
ChangDebtChoice	ChangDebtChoice (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
ChangExternalFunding	ChangExternalFunding (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
DailyVola	Build Volatility from raw CRSP daily stock files via CRSPEngine. *Formula:* CRSP/RET (see module: src/f1d/shared/variables/volatility.py)	CRSP/RET	dwz2021
DebtToCapital	Build DebtToCapital = total debt / (equity + total debt). *Formula:* Compustat/dlcq,dlttq,seqq (see module: src/f1d/shared/variables/debt_to_capital.py)	Compustat/dlcq,dlttq,seqq	rz1995
DivDummy	Build DivDummy = (dvy > 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvy>0 (see module: src/f1d/shared/variables/dividend_payer.py)	Compustat/dvy>0	Bates et al. (2009) (Section IV, p.2000; dividend payout dummy based on common dividends)
DivPayerQ	Build DivPayerQ = (dvpsxq > 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvpsxq>0 (quarterly ex-date) (see module: src/f1d/shared/variables/dividend_payer_quarterly.py)	Compustat/dvpsxq>0	Hoberg & Prabhala (2009, RFS) — Compustat item 26 (dvpsx) analog

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Name	Description / Formula	Source	Reference
EarnVol	Builder for Earnings Volatility (earnings_volatility) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/earnings_volatility.py)	Compustat	Duong et al. (2025) (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)
EquityDelayCon	Hoberg-Maksimovic (2015) equity-market financing constraint index (H22). *Formula:* Hoberg-Maksimovic (2015) firm-year equity-market financing constraint measure. Higher values = more constrained firm. Panel column is lowercase ‘equitydelaycon‘; CamelCase name is the spec convention. See docs/provenance/h22.md for merge details.	inputs/Hoberg_Maksimovic/	hm2015
FirmMat	Builder for Firm Maturity (firm_maturity) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/firm_maturity.py)	Compustat	Biddle et al. (2009) (Appendix A, p.130; years since first CRSP appearance); alt Leary and Roberts (2010) (App.~A p.354)
Leverage	Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only). *Formula:* Compustat/dlcq,dlttq,atq (see module: src/f1d/shared/variables/book_lev.py)	Compustat/dlcq,dlttq,atq	bks2009, rz1995
lnAssets	Build Size = ln(atq) from raw Compustat quarterly data. *Formula:* Compustat/atq (see module: src/f1d/shared/variables/size.py)	Compustat/atq	dwz2021
Loss	Builder for Loss (H5 Control Variable). *Formula:* Compustat ibq < 0 (see module: src/f1d/shared/variables/loss_dummy.py)	Compustatibq<0	Biddle et al. (2009) (Appendix A, p.131; indicator for negative net income before extraordinary items)
PayoutRatio_q	Build PayoutRatio_q = (dvpspq × cshoq) / ibq from Compustat. *Formula:* Compustat/(dvpspq*cshoq)/ibq (quarterly) (see module: src/f1d/shared/variables/payout_ratio_quarterly.py)	Compustat/	dds2004
PRisk	Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter). *Formula:* Hassan et al. (2019) quarterly PRisk (see module: src/f1d/shared/variables/prisk_q.py)	Hassanetal.	hhlt2019
RDSales	Build RDSales = xrdq / atq from raw Compustat quarterly data. *Formula:* Compustat/xrdq,atq (see module: src/f1d/shared/variables/rd_intensity.py)	Compustat/xrdq,atq	bks2009, jjl2021
RepurchaseIntensity	Build RepurchaseIntensity = quarterly_prstkcy / lagged_atq from Compustat. *Formula:* Compustat/quarterly_prstkcy/lagged_atq (see module: src/f1d/shared/variables/repurchase_intensity.py)	Compustat/quarterly_prstkcy/lagged_atq	Thesis Advisor Suggestion
ROA	Build ROA = iby_annual / avg_assets from Compustat quarterly data. *Formula:* Compustat/iby,atq (see module: src/f1d/shared/variables/roa.py)	Compustat/iby,atq	Wang (2020), dwz2021
SalesGrowth	Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel). *Formula:* Compustat/saley_Q4_YoY (see module: src/f1d/shared/variables/sales_growth.py)	Compustat/saley_Q4_YoY	Biddle et al. (2009) (Section 3.2, p.117; pct change in sales)
sCFO	Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey. *Formula:* Compustat/oancfy/atq_{t-1} rolling std (see module: src/f1d/shared/variables/ocf_volatility.py)	Compustat/oancfy/atq_{t-1}rollingstd	Biddle et al. (2009) (Appendix A, p.130; 5-year rolling std of CFO/avg assets)
StockPrice	Compute stock price at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/stock_price.py)	CRSPviaget_raw_daily_data	Amiram et al. (2016) (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)

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Name	Description / Formula	Source	Reference
`SurpDec`	Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter. *Formula:* Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009).	.../variables/earnings_surprise.py	Dzielínski et al. (2021) (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)
`TobinsQ`	Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data. *Formula:* Compustat/(atq+cshoq*prccq-ceqq)/atq (see module: src/f1d/shared/variables/tobins_q.py)	Compustat/	Bates, Kahle & Stulz (2009, JF)
`Turnover`	Compute share turnover at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/turnover.py)	CRSPviaget_raw_daily_data	Amihud (2002) (Section 2.1, p.33; alt rolling-window def in Chang et al. (2006) Appendix B p.3045)

A.4 Econometric / Indicator Variables

Name	Description / Formula	Source	Reference
`BelowIG`	Below-IG credit rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm's S&P long-term issuer credit rating is below investment grade (BB+ and below) at call date; 0 otherwise. Rated firms only (Unrated firms get separate indicator).	runtime	thesis (H1.2 rating subsample)
`HighTSIMM`	High product-market-similarity indicator (H1.1). *Formula:* Binary = 1 if firm-year TotalSimilarity (HP 2016 product market similarity) is above the within-year median; 0 otherwise.	runtime	hp2016 (TSIMM construction); thesis (H1.1 split convention)
`Lagged_DV`	Generic lagged-DV control (1Q lag of suite-specific DV). *Formula:* 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.	runtime	thesis convention (per feedback_lagged_dv.md)
`Unrated`	No-rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm has no S&P rating at call date; 0 otherwise.	runtime	thesis (H1.2 rating subsample)
`z_log_TotalSimilarity`	z-scored log of HP 2016 TotalSimilarity (continuous moderator, H1.1). *Formula:* z-score of log(1 + TotalSimilarity) computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities $\times$ 100).	runtime	hp2016

A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

Name	Description / Formula	Source	Reference
`AbsSurpDec`	Absolute earnings surprise quintile magnitude. *Formula:* abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5.	runtime	thesis (magnitude control independent of sign)
`BGTAvg_Amihud_lead1`	1Q lead of BGTAvg_Amihud. *Formula:* 1-quarter lead of BGTAvg_Amihud	runtime	bgt2018 (window); F1D extension (symmetric average shape)

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Name	Description / Formula	Source	Reference
BGTAvg_Spread_lead1	1Q lead of BGTAvg_Spread. *Formula:* 1-quarter lead of BGTAvg_Spread	runtime	bgt2018 (window); F1D extension (symmetric average shape)
BGTDelta_Amihud_lead1	1Q lead of BGTDelta_Amihud. *Formula:* 1-quarter lead of BGTDelta_Amihud	runtime	bgt2018 (window); F1D extension (delta shape)
BGTDelta_Spread_lead1	1Q lead of BGTDelta_Spread. *Formula:* 1-quarter lead of BGTDelta_Spread	runtime	bgt2018 (window); F1D extension (delta shape)
BGTLevel_Amihud_lead1	1Q lead of BGTLevel_Amihud. *Formula:* 1-quarter lead of BGTLevel_Amihud	runtime	bgt2018
BGTLevel_Spread_lead1	1Q lead of BGTLevel_Spread. *Formula:* 1-quarter lead of BGTLevel_Spread	runtime	bgt2018, lee2016
Capex_lead	1Q lead of Capex. *Formula:* 1-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead2	2Q lead of Capex. *Formula:* 2-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead3	3Q lead of Capex. *Formula:* 3-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead4	4Q lead of Capex. *Formula:* 4-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
CashRatio_lead	1Q lead of CashRatio. *Formula:* 1-quarter lead of CashRatio	runtime	see base var
ChangDebtChoice_lead	1Q lead of ChangDebtChoice. *Formula:* 1-quarter lead of ChangDebtChoice	runtime	cdh2006
ChangExternalFunding_lead	1Q lead of ChangExternalFunding. *Formula:* 1-quarter lead of ChangExternalFunding	runtime	cdh2006
DebtToCapital_lead	1Q lead of DebtToCapital. *Formula:* 1-quarter lead of DebtToCapital	runtime	rz1995
DeltaILLIQ_lead1	1Q lead of DeltaILLIQ. *Formula:* 1-quarter lead of DeltaILLIQ	runtime	amihud02
DISP_lag	1Q lag of DISP. *Formula:* 1-quarter lag of DISP	runtime	Wang (2020)
DISP_lead	1Q lead of DISP. *Formula:* 1-quarter lead of DISP	runtime	Wang (2020)
DivPayerQ_lead1	1Q lead of DivPayerQ. *Formula:* 1-quarter lead of DivPayerQ	runtime	Hoberg & Prabhala (2009, RFS) – primary; Chetty & Saez (2005, QJE) – secondary (firm-quarter frequency precedent)
DSPREAD_lead1	1Q lead of DSPREAD. *Formula:* 1-quarter lead of DSPREAD	runtime	lee2016

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Name	Description / Formula	Source	Reference
`EquityDelayCon_lead`	1Q lead of EquityDelayCon. *Formula:* 1-quarter lead of EquityDelayCon	runtime	hm2015
`Leverage_lead`	1Q lead of Leverage. *Formula:* 1-quarter lead of Leverage	runtime	bks2009, rz1995
`PayoutRatio_q_lead_qtr`	1Q lead of PayoutRatio_q (qtr-aligned). *Formula:* 1-quarter lead of PayoutRatio_q (calendar-quarter aligned)	runtime	dds2004
`PostCallAmihud`	Panel-time post-call Amihud illiquidity level (H7b). *Formula:* Panel-time mean daily Amihud illiquidity over the post-call period (call quarter + N following quarters per H7b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H7b panel-time post-call construction)
`PostCallAmihud_lead1`	1Q lead of PostCallAmihud. *Formula:* 1-quarter lead of PostCallAmihud	runtime	thesis (H7b panel-time post-call construction)
`PostCallSpread`	Panel-time post-call bid-ask spread level (H14b). *Formula:* Panel-time mean daily relative bid-ask spread over the post-call period (call quarter + N following quarters per H14b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H14b panel-time post-call construction)
`PostCallSpread_lead1`	1Q lead of PostCallSpread. *Formula:* 1-quarter lead of PostCallSpread	runtime	thesis (H14b panel-time post-call construction)
`PRisk_lag`	1Q lag of PRisk. *Formula:* 1-quarter lag of PRisk	runtime	hhlt2019
`PRisk_lag2`	2Q lag of PRisk. *Formula:* 2-quarter lag of PRisk	runtime	hhlt2019
`RDSales_lead`	1Q lead of RDSales. *Formula:* 1-quarter lead of RDSales	runtime	bks2009, jjl2021
`RepurchaseIntensity_lead_qtr`	1Q lead of RepurchaseIntensity (qtr-aligned). *Formula:* 1-quarter lead of RepurchaseIntensity (calendar-quarter aligned)	runtime	Thesis Advisor Suggestion
`UncAnsMgr_c`	Mean-centered UncAnsMgr. *Formula:* UncAnsMgr minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)
`UncAnsMgr_c_x_BelowIG`	Interaction term: UncAnsMgr_c x BelowIG. *Formula:* Product: UncAnsMgr_c * BelowIG	runtime	thesis (interaction term)
`UncAnsMgr_c_x_HighTSIMM`	Interaction term: UncAnsMgr_c x HighTSIMM. *Formula:* Product: UncAnsMgr_c * HighTSIMM	runtime	thesis (interaction term)
`UncAnsMgr_c_x_IG`	Interaction term: UncAnsMgr_c x IG. *Formula:* Product: UncAnsMgr_c * IG	runtime	thesis (interaction term)
`UncAnsMgr_c_x_Unrated`	Interaction term: UncAnsMgr_c x Unrated. *Formula:* Product: UncAnsMgr_c * Unrated	runtime	thesis (interaction term)
`UncAnsMgr_c_x_zlogTSIMM`	Interaction term: UncAnsMgr_c x zlogTSIMM. *Formula:* Product: UncAnsMgr_c * zlogTSIMM	runtime	thesis (interaction term)

A.6 Stage 9

Name	Description / Formula	Source	Reference
CCCL	SEC comment letter received (binary, forward-looking) *Formula:* 1 if SEC comment letter received in next quarter, 0 otherwise	inputs/EDGAR_CommentLetters	cdm2013
DeltaILLIQ	Change in Amihud illiquidity around earnings call *Formula:* PostCallILLIQ - PreCallILLIQ over [+1,+3] vs [-3,-1] days	inputs/CRSP_DSF	amihud02
DISP	Analyst forecast dispersion (Wang 2020) *Formula:* SD(latest analyst EPS forecasts in T-31..T-1 window) / prccq_prior	inputs/tr_ibes	Wang (2020)
DSPREAD	Change in closing bid-ask spread around earnings call *Formula:* mean(RelSpread[+1,+3]) - mean(RelSpread[-3,-1]) where RelSpread = 2*(ASK-BID)/(ASK+BID)	inputs/CRSP_DSF	lee2016
GEPU_log	Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-$ GDP weights) *Formula:* log(GEPU_current) — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/Global	Davis (2016)
GPR_log	Caldara & Iacoviello (2022, AER) headline Geopolitical Risk index (log-transformed, 10-newspaper recent version) *Formula:* log(GPR) — recent headline Geopolitical Risk index, matched by calendar month of call start_date	inputs/matteoiacoviello/GPR_Global	ci2022
SEC_Letters_fwd	SEC comment letter count (forward-looking) *Formula:* Count of SEC-originated letters (UPLOAD type) in next calendar quarter	inputs/EDGAR_CommentLetters	cdm2013 (thesis extension to count)
TotalSimilarity	Hoberg-Phillips total product similarity *Formula:* Sum of pairwise cosine similarities from 10-K product descriptions	inputs/TNIC	hp2016
US_EPU_log	Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed) *Formula:* log(News_Based_Policy_Uncert_Index) matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/US	bbd2016

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